Unit 7, Lesson 4: Graphing Lines from Slope-Intercept Form

Benchmark

MA.8.AR.3.4 Given a mathematical or real-world context, graph a two-variable linear equation from a written description, a table, or an equation in slope-intercept form.

Learning Targets

- ☐ I can graph a two-variable linear equation from an equation in slope-intercept form.
- ☐ I can graph a two-variable linear equation from a table of values.

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7.4.1 Warm-Up: Maximizing Slope

1. Place a digit in each blue box to find a point that creates the steepest slope for a line that passes through the point $(3,5)$. Use only digits 1 to 9, at most one time each.

$(\square, \square)$

7.4.2 Guided Instruction: Graphing Two-Variable Linear Equations

The graph of a linear relationship can be created from a two-variable linear equation.

When given the equation in slope-intercept form, start by determining the slope and the y-intercept of the line.

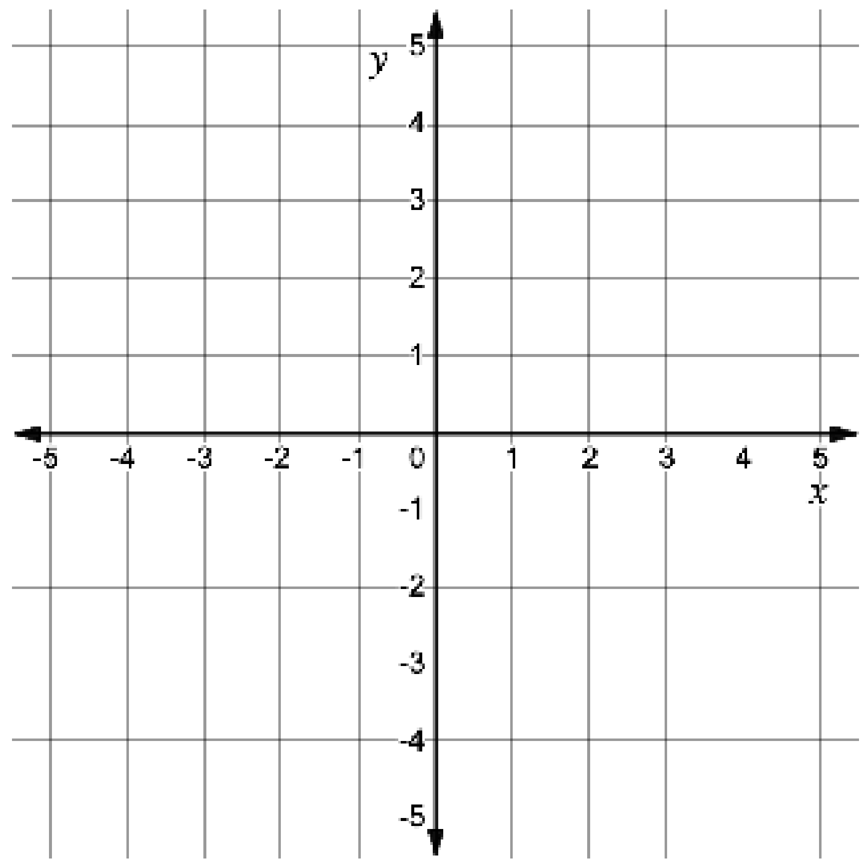
1. Complete the table.

Equation	Slope	y-intercept
$y = -\frac{3}{4}x - 1$		

Recall that the y-intercept is the value of y when $x = 0$, at the point where the line intersects the y-axis.

2. Write the y-intercept as a coordinate pair. (____, ____)

3. Plot the y-intercept on the graph.



4. Use the slope to find two other points on the line.

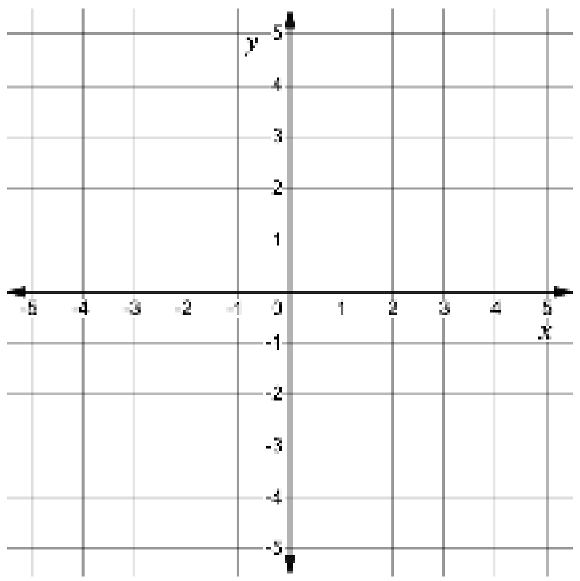
5. Use a straightedge to draw the line that represents the linear relationship.

Two-variable linear relationships can also be written in other forms. When given the equation in another form, start graphing by determining points on the line using the equation.

6. Complete the table of values using the equation.

Equation	Table of Values	
	x	y
$-2x + 3y = 9$		0
	0	
	3	

7. Plot the ordered pairs from the table of values on the graph. Then, use a straightedge to draw the line that represents the linear relationship.



8. Ana says that to graph a linear relationship, only two points on the line are needed.

Discuss with your partner whether you agree or disagree with Ana and explain why.

9. Complete the statements.

The slope of the line between any two points in a linear relationship
o varies
o is the same
because the rate of change, $\frac{\text{change in } y}{\text{change in } x}$, is
o constant.
o always changing.
Therefore,
o only two points
o more than two points
are needed to graph the line using a straightedge.

When given the equation of a line in slope-intercept form, two points on the line can be determined using the y-intercept and slope. Start by graphing the y-intercept. Then, use the slope to determine a second point to graph from there.

When given the equation of a line in a form other than slope-intercept, two points on the line can be determined by using substitution to create a table of values.

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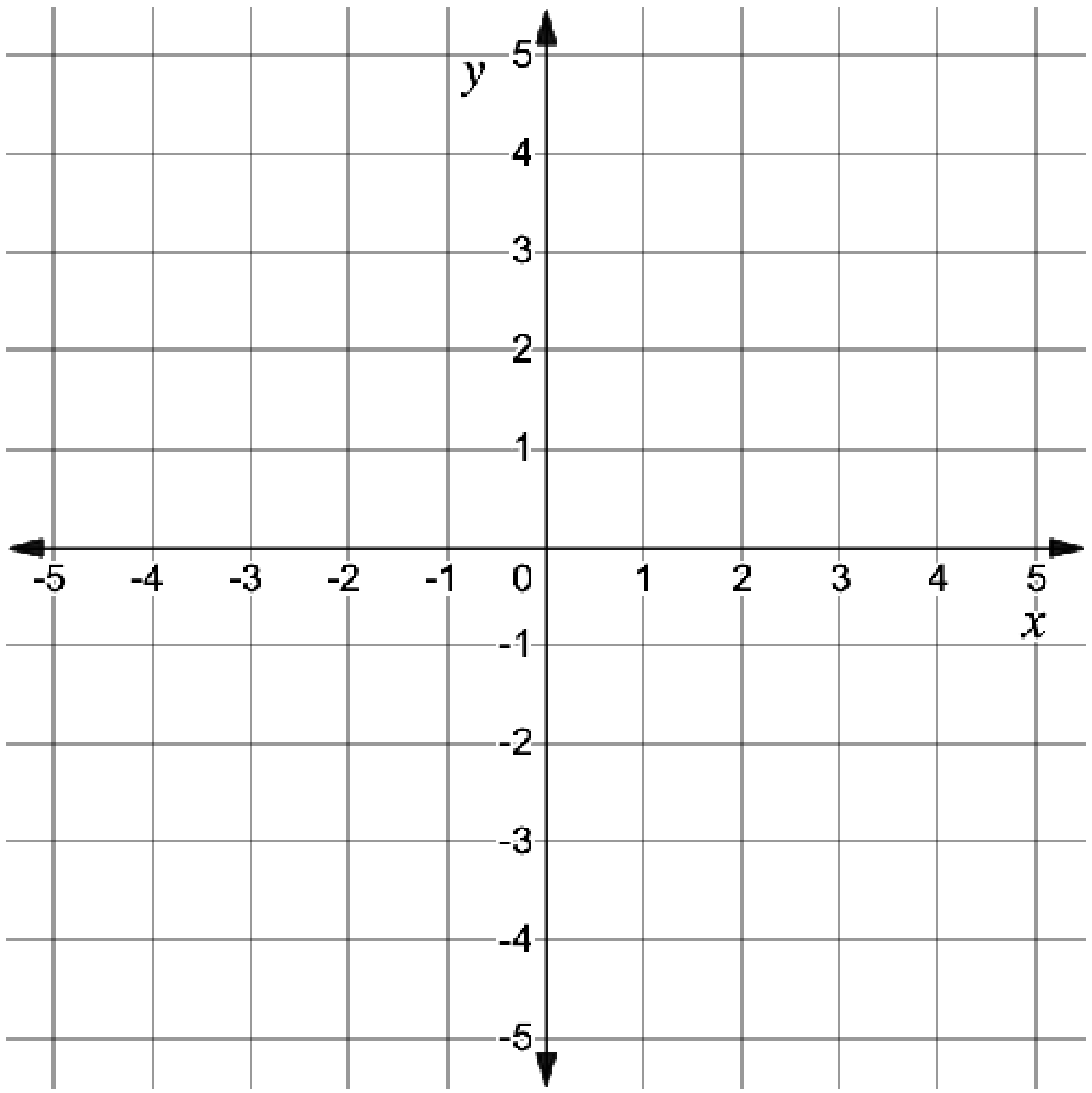
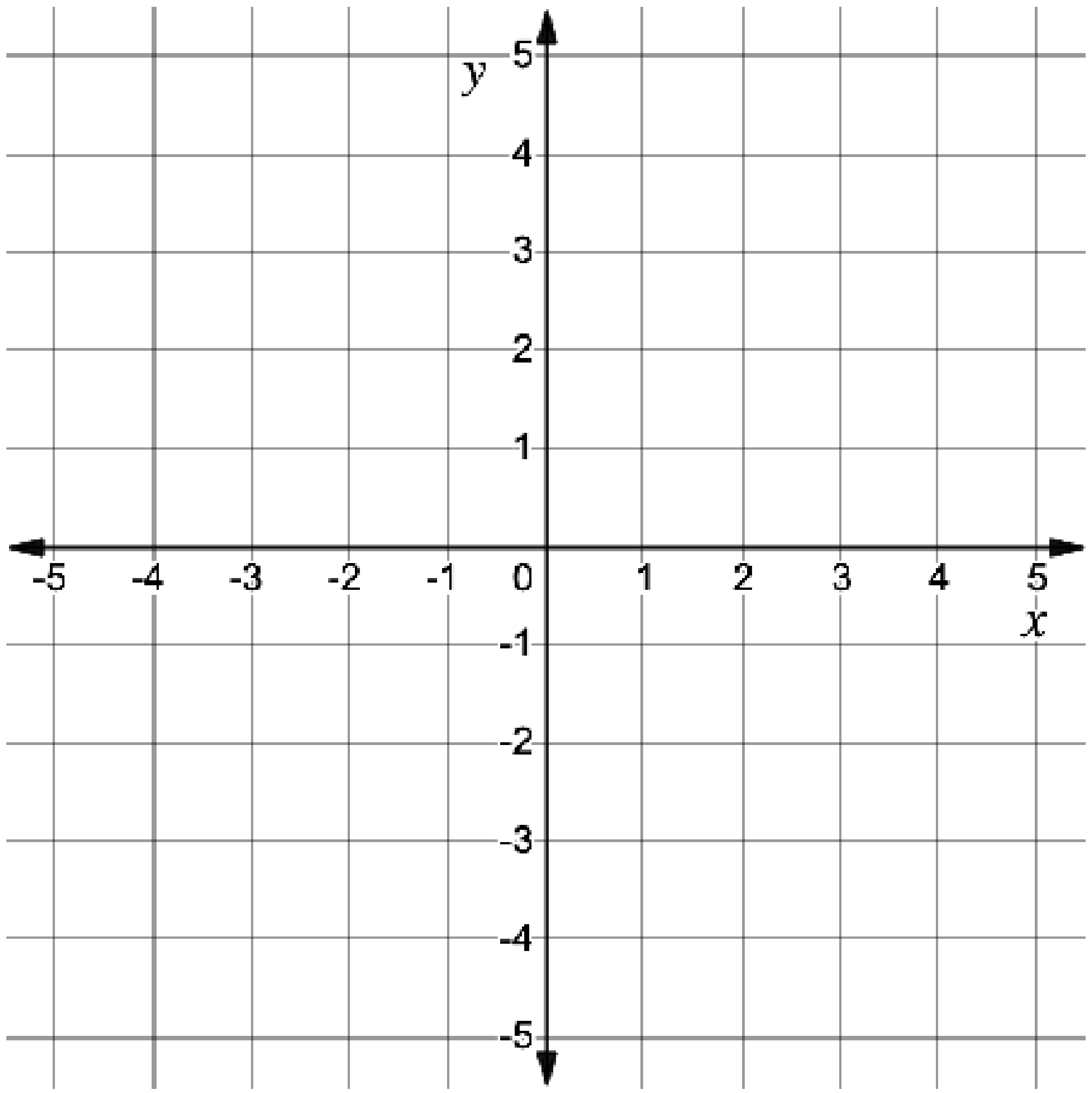
7.4.3 Try It: Graphing Two-Variable Linear Equations

1. Graph the equation $y = -\frac{5}{2}x - 1$.



2. Graph the equation $y = 3x - 5$.

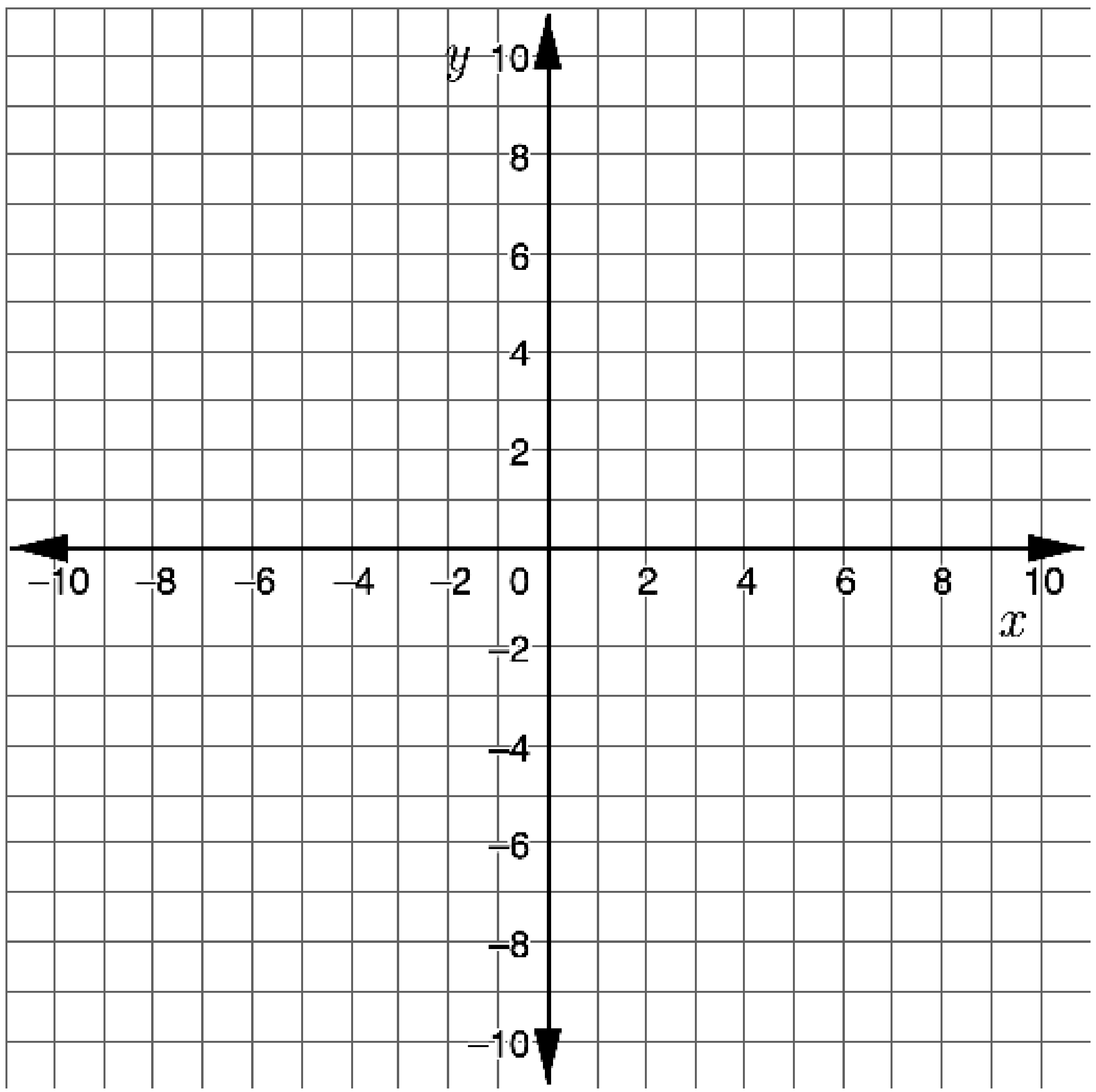


3. Consider the equation $x + 2y = 8$.

a. Use the equation to complete the table of values.

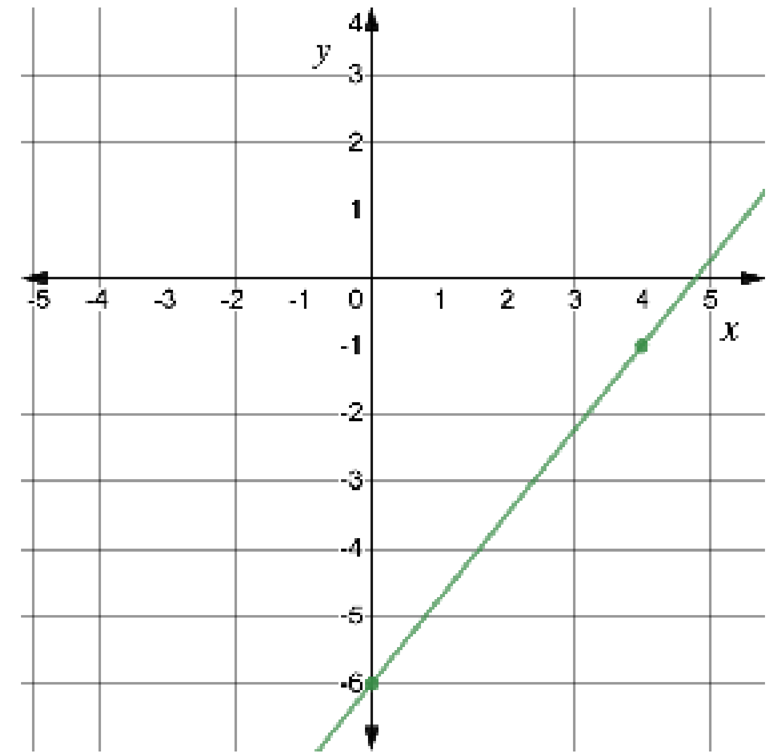
x	y

b. Use the table of values to graph the equation.



7.4.4 Collaboration: Error Analysis

Nico graphed the equation $y = \frac{4}{5}x - 6$. His graph is shown.



1. Nico made an error when he graphed the line $y = \frac{4}{5}x - 6$. Work with your partner to find the error.

2. Help Nico understand how he can learn from his mistake. Write a brief note to Nico explaining what he did correctly, what his mistake was, and how he can learn from it.

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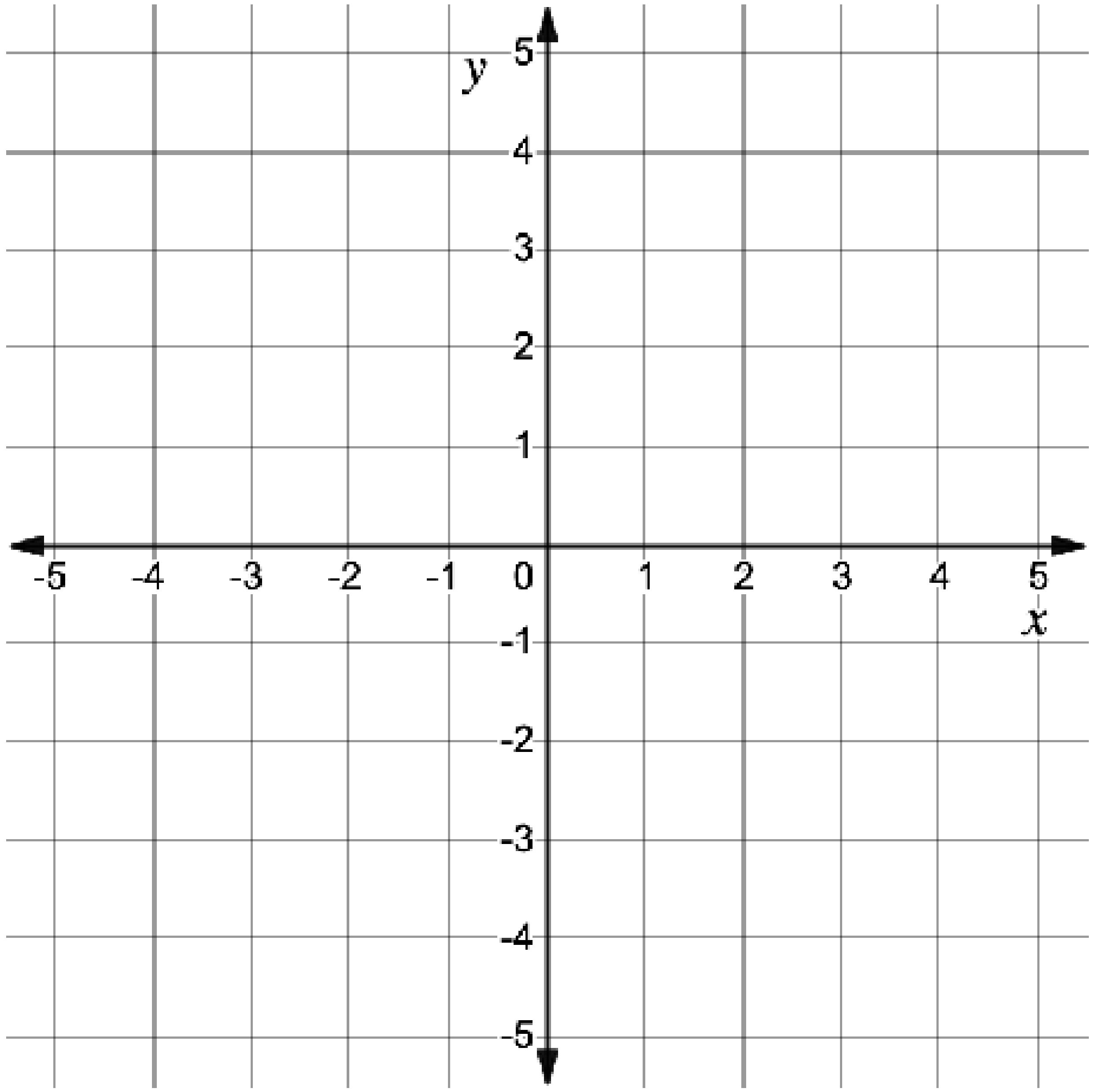
7.4.5 Your Turn!

1. Graph the equation $y = \frac{3}{4}x - 2$.

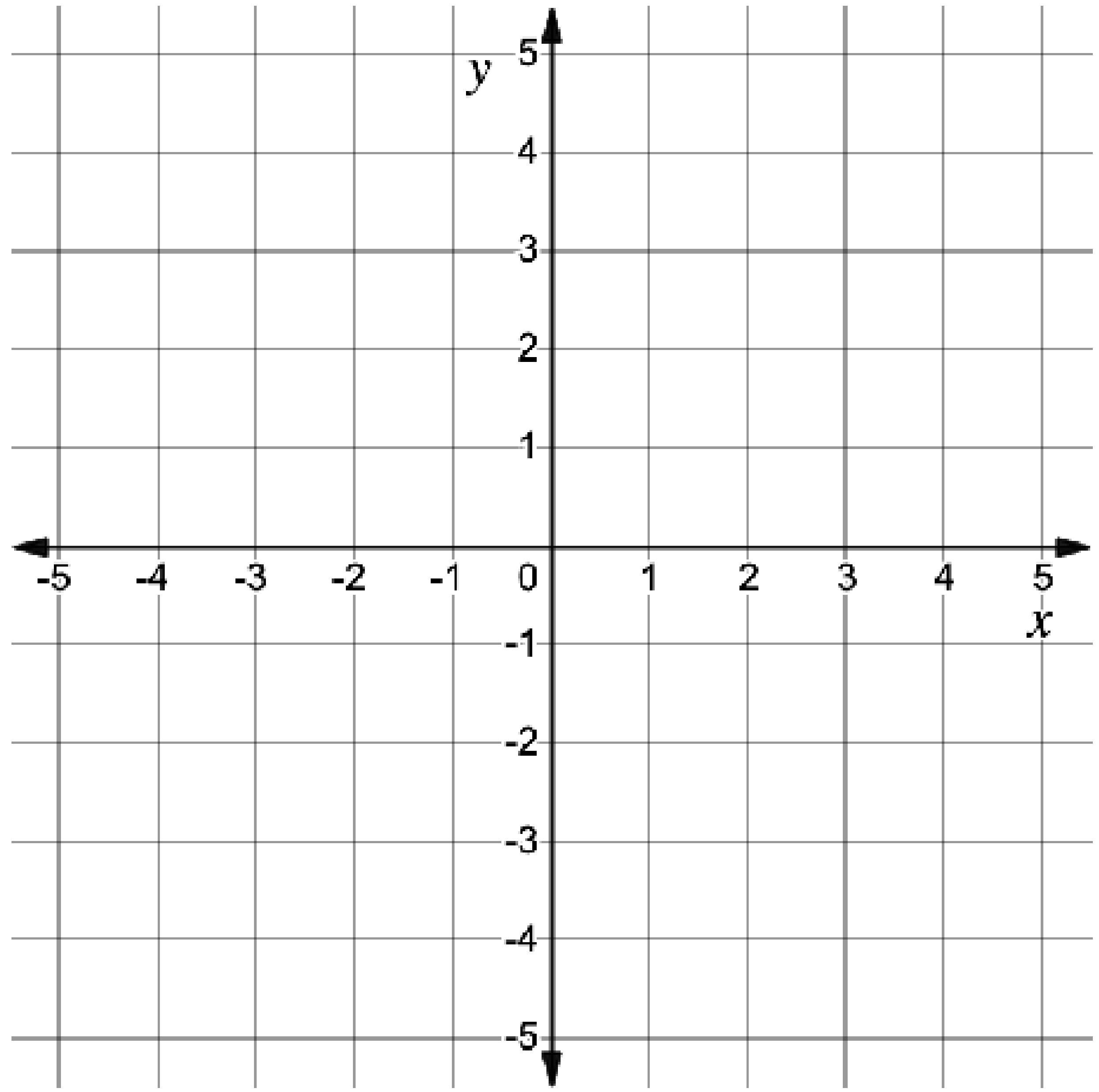


2. Consider the equation $5x - 2y = 6$.

a. Use the equation to complete the table of values.

x	y

b. Use the table of values to graph the equation $5x - 2y = 6$.



c. What is the slope of the line?

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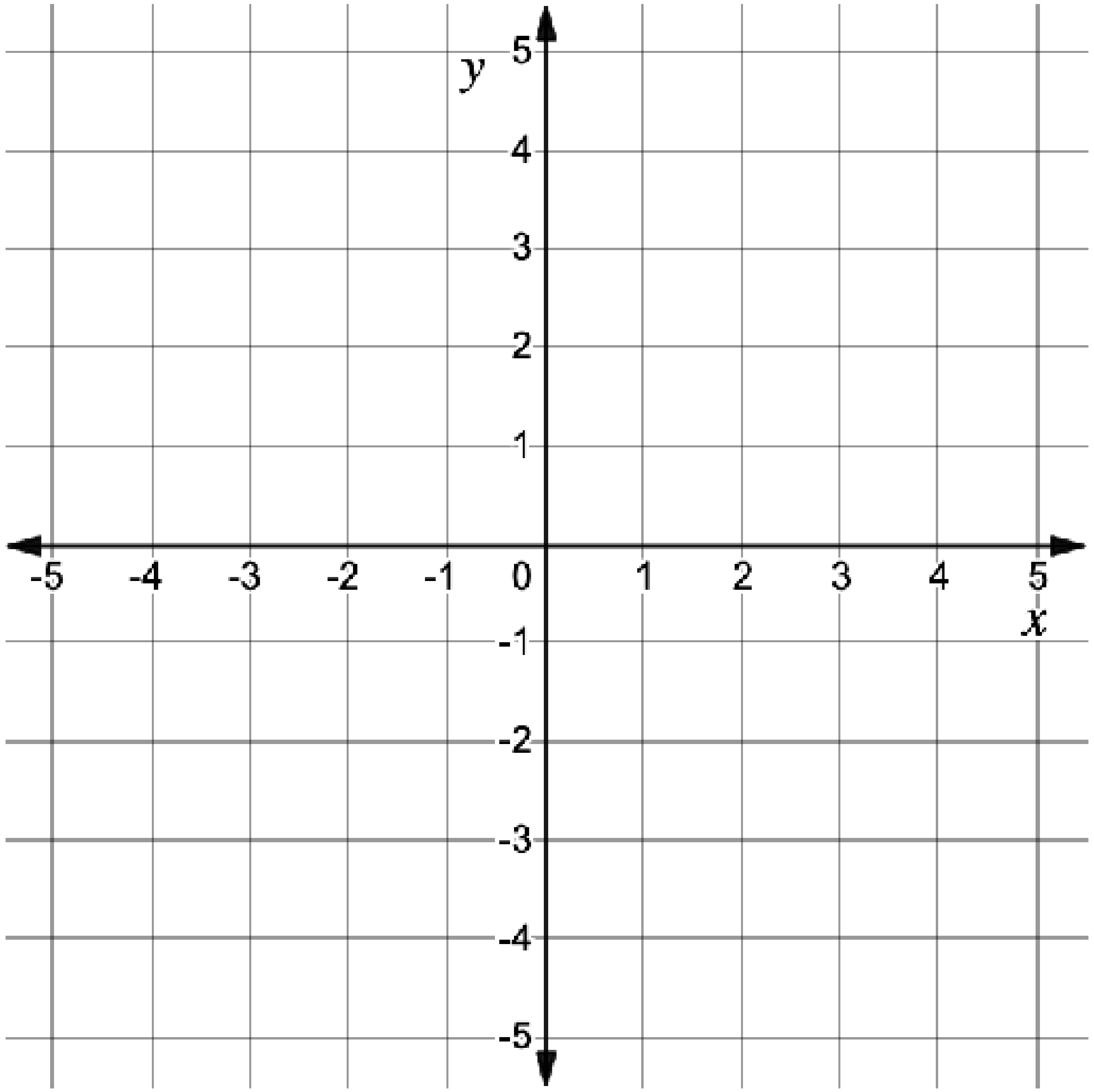
3. Graph the equation $y = -\frac{1}{5}x + 1$.



7.4.6 Wrap-Up: Reflection

1. Complete the learning pyramid based on today's lesson.

One question I would like to have answered ...

One mistake I learned from today ...

One thing I am not sure about ... YET!

One thing I am confident I know is ...